Derivation and Implementation Notes

Woloshyn — Demonstrating Quantum Computing with the Quark Model
arXiv:2301.10828v2

Companion document for the `quarksim` codebase

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1 The Physics Problem

1.1 Charmonium with Spin-Dependent Interactions

This paper extends the quantum computing approach to charmonium by including **spin-dependent interactions**, which break the degeneracy between spin-singlet and spin-triplet states. Three channels are studied:

- $^1S_0$ (η_c): orbital $l = 0$, spin singlet ($\vec{S}_c \cdot \vec{S}_{\bar{c}} = -3/4$)
- $^3S_1$ (J/ψ): orbital $l = 0$, spin triplet ($\vec{S}_c \cdot \vec{S}_{\bar{c}} = +1/4$)
- $^1P_1$ (h_c): orbital $l = 1$, spin singlet

1.2 The Cornell Potential with Spin Contact Term

The potential between the charm quark and anti-charm quark is:

$$V(r) = -\frac{a}{r} + b\,r + V_s(r)\,\vec{S}_c \cdot \vec{S}_{\bar{c}} \tag{1}$$

where $a = 4\alpha_s/3$ is the color Coulomb coupling, b is the string tension, and $V_s(r)$ is the spin-dependent contact potential:

$$V_s(r) = \frac{32\pi\alpha_s}{9\,m_c^2}\,\delta(r) \tag{2}$$

The delta function is smeared with a Gaussian:

$$\delta(r) = \left(\frac{\sigma}{\sqrt{\pi}}\right)^3 e^{-\sigma^2 r^2} \tag{3}$$

Note that spin-orbit and tensor interactions are omitted as they are not relevant for the states considered here.

Contrast with Gallimore: The Gallimore implementation uses a spin-averaged potential ($V = -\kappa/r + \sigma r$) and considers only spin-averaged S-wave states. Woloshyn's inclusion of the contact term enables the study of the physical mass splitting between η_c (2982 MeV) and J/ψ (3097 MeV), a difference of 115 MeV.

1.3 Parameters

The parameters from Barnes *et al.* (Table 1 of the paper):

Symbol	Value	Description
α_s	0.5461	Strong coupling constant
b	0.1425 GeV^2	String tension
m_c	1.4794 GeV	Charm quark mass
σ	1.0946 GeV	Gaussian smearing parameter
ω	1.2 fm^{-1}	HO frequency (chosen from Fig. 1)

Units: Unlike Gallimore (which works in MeV), all energies here are in fm^{-1}. The conversion factor is $\hbar c = 197.326$ MeV·fm, so 1 fm^{-1} = 197.3 MeV. The meson mass is $M = E + 2m_c$, where E is the energy eigenvalue.

Code: `hamiltonian.py`, constants in `CHARMONIUM_PARAMS`.

2 Harmonic Oscillator Basis

2.1 Basis States

The Hamiltonian for the 3D harmonic oscillator is (Eq. 2):

$$H_{\mathrm{HO}} = -\frac{\vec{\nabla}^2}{2\mu} + \frac{1}{2}\mu\omega^2 r^2 \tag{4}$$

The quark model Hamiltonian is rewritten as (Eq. 3):

$$H_{\mathrm{qm}} = H_{\mathrm{HO}} + V(r) - \frac{1}{2}\mu\omega^2 r^2 \tag{5}$$

This decomposition is convenient because the matrix elements of H_{HO} are trivially diagonal, and $V(r) - \frac{1}{2}\mu\omega^2 r^2$ contains the "residual" interaction.

The HO basis wavefunctions (unnormalized) are:

$$r^l\, e^{-\frac{1}{2}\nu r^2}\, L_n^{l+1/2}(\nu r^2)\, Y_{lm}(\theta, \phi) \tag{6}$$

where $\nu = \mu\omega$, $L_n^{l+1/2}$ is a generalized Laguerre polynomial, and the eigenenergy is $\omega(2n+l+3/2)$ with $\hbar = 1$.

2.2 Truncation to $N = 4$ States

The basis is truncated to $N = 4$ radial states ($n = 0, 1, 2, 3$) for each channel. This gives a 4×4 Hamiltonian matrix encodable in **2 qubits** — half the qubits needed for Gallimore's 3-orbital system.

The oscillator parameter $\omega = 1.2$ fm^{-1} is chosen as a compromise: small ω gives better convergence for low-lying states (closer to the exact Schrödinger solution) but worse for higher states. The paper's Fig. 1 shows the eigenvalues vs. ω, demonstrating the variational nature of the truncation.

Code: The matrix is built in `hamiltonian.get_matrix(channel)` using the paper's explicit matrices, or computed from scratch via `hamiltonian.compute_matrix(channel)`.

3 Matrix Elements

3.1 Structure

The matrix elements of H_{qm} in the HO basis are:

$$\langle m|H_{\mathrm{qm}}|n\rangle = \omega\big(2n + l + \tfrac{3}{2}\big)\,\delta_{mn} + \langle m|V(r) - \tfrac{1}{2}\mu\omega^2 r^2|n\rangle \tag{7}$$

The first term is the diagonal HO eigenvalue. The second term contains:

- $-a\,\langle m|r^{-1}|n\rangle$: Coulomb integral (hypergeometric functions, same as Gallimore)
- $b\,\langle m|r|n\rangle$: Linear confinement integral (hypergeometric functions)
- $-\frac{1}{2}\mu\omega^2\,\langle m|r^2|n\rangle$: HO potential subtraction (tridiagonal, closed form)
- $V_s \cdot S \cdot S \cdot \langle m|\delta(r)|n\rangle$: Spin contact integral (numerical quadrature, S-waves only)

3.2 Explicit Matrices

The $^1S_0$ Hamiltonian matrix at $\omega = 1.2$ fm^{-1} (Eq. 4):

$$H_{^1S_0} = \begin{pmatrix} 0.9431 & -0.8733 & -0.7690 & -0.5601 \\ -0.8733 & 3.3365 & -0.5646 & -0.8648 \\ -0.7690 & -0.5646 & 5.4382 & -0.1566 \\ -0.5601 & -0.8648 & -0.1566 & 7.3451 \end{pmatrix} \tag{8}$$

The $^3S_1$ matrix (Eq. 15 in the Appendix):

$$H_{^3S_1} = \begin{pmatrix} 1.0946 & -0.7114 & -0.6111 & -0.4112 \\ -0.7114 & 3.5406 & -0.3910 & -0.6989 \\ -0.6111 & -0.3910 & 5.6122 & -0.0119 \\ -0.4112 & -0.6989 & -0.0119 & 7.5104 \end{pmatrix} \tag{9}$$

The difference between $^1S_0$ and $^3S_1$ on the diagonal is due to the spin contact interaction: $\Delta H_{nn} = V_s \cdot [\frac{1}{4} - (-\frac{3}{4})] \cdot \langle n|\delta(r)|n\rangle$. For $n = 0$: $\Delta = 1.0946 - 0.9431 = 0.1515\,\text{fm}^{-1}$, corresponding to 30 MeV of hyperfine splitting per level.

3.3 Eigenvalues (Table 3)

State	$^1S_0$ (fm^{-1})	$^3S_1$ (fm^{-1})	$^1P_1$ (fm^{-1})
1	0.395	0.753	2.783
2	3.506	3.634	4.875
3	5.664	5.723	6.765
4	7.546	7.648	8.767

These are the target values for the VQITE algorithm.

Code: Eigenvalues via `numpy.linalg.eigvalsh(get_matrix(channel))`.

4 Hamiltonian in Pauli Form (2 Qubits)

4.1 Direct Encoding

With 4 basis states encoded in 2 qubits, the mapping is direct:

$$|00\rangle \leftrightarrow \text{HO state } n = 0$$
$$|01\rangle \leftrightarrow \text{HO state } n = 1$$
$$|10\rangle \leftrightarrow \text{HO state } n = 2$$
$$|11\rangle \leftrightarrow \text{HO state } n = 3$$

Key difference from Gallimore: In the Jordan-Wigner encoding (3 qubits), only 3 of the $2^3 = 8$ computational basis states are physical (the 1-particle sector). Here, *all* $2^2 = 4$ states are physical. This means:

- No need to project onto a physical subspace.
- The full 4×4 Pauli matrix equals the physical Hamiltonian directly.
- Fewer qubits (2 vs. 3) for more basis states (4 vs. 3).

4.2 Pauli Decomposition

A general 4×4 real symmetric matrix decomposes into 10 Pauli terms (the 6 imaginary operators XY, YX, ZY, YZ, IY, YI vanish for real matrices):

$$H = c_0\, II + c_1\, Z_0 + c_2\, Z_1 + c_3\, Z_0Z_1 + c_4\, X_0 + c_5\, X_1 + c_6\, Z_0X_1 + c_7\, X_0Z_1 + c_8\, X_0X_1 + c_9\, Y_0Y_1 \tag{10}$$

The coefficients are computed as:

$$c_k = \frac{1}{4}\text{Tr}(H \cdot P_k) \tag{11}$$

4.3 Pauli Coefficients

Term	$^1S_0$	$^3S_1$	$^1P_1$
c_0 (II)	4.273	4.439	5.798
c_1 (Z_0)	−2.119	−2.122	−1.910
c_2 (Z_1)	−1.082	−1.086	−0.964
c_3 (Z_0Z_1)	−0.129	−0.137	−0.067
c_4 (X_0)	−0.817	−0.655	−0.446
c_5 (X_1)	−0.515	−0.350	0.083
c_6 (Z_0X_1)	−0.358	−0.361	−0.323
c_7 (X_0Z_1)	0.048	0.044	−0.064
c_8 (X_0X_1)	−0.562	−0.401	−0.095
c_9 (Y_0Y_1)	−0.002	0.010	0.133

Note on qubit ordering: The paper uses PennyLane (big-endian: qubit 0 = MSB), while our Qiskit implementation uses little-endian (qubit 0 = LSB). The paper's Z_0 corresponds to Qiskit's `'ZI'`, and Z_1 to `'IZ'`. The coefficients are physically identical; only the labeling convention differs.

Code: `hamiltonian.build_pauli_hamiltonian(channel)` decomposes via `matrix_to_pauli(H)`. Validation via `validate_pauli_hamiltonian()` confirms the 4×4 Pauli matrix reconstructs the original matrix exactly.

5 Variational Quantum Imaginary Time Evolution

5.1 Imaginary Time Evolution

For a Hamiltonian H with eigenvalues $\{\lambda_i\}$ and eigenstates $\{|\phi_i\rangle\}$, the imaginary time evolution of an initial state $|\psi_0\rangle$ is (Eq. 5):

$$e^{-H\tau}\,|\psi_0\rangle = \sum_i e^{-\lambda_i \tau}\,\langle\phi_i|\psi_0|\phi_i|\psi_0\rangle\,|\phi_i\rangle \tag{12}$$

For large τ, the lowest-energy component dominates. However, this operation is **non-unitary** and cannot be directly implemented on a quantum computer.

5.2 The Variational Approach

We parametrize the state as $\left|\psi(\vec{\theta})\right\rangle = U(\vec{\theta})\,|0\rangle$ and seek unitary updates that approximate imaginary time evolution. The variational principle (Eqs. 7–8) leads to the evolution equation (Eq. 9):

$$\boxed{\sum_j A_{ij}\,\dot{\theta}_j = C_i} \tag{13}$$

where:

$$C_i = -\frac{\partial E_\tau}{\partial \theta_i} \tag{14}$$

$$A_{ij} = -\frac{\partial^2 \left\langle \psi(\tilde{\theta})|\psi(\vec{\theta})\middle|\psi(\tilde{\theta})|\psi(\vec{\theta})\right\rangle}{\partial\theta_i\,\partial\theta_j} \tag{15}$$

Here $E_\tau = \left\langle \psi(\vec{\theta}(\tau))\middle|H\middle|\psi(\vec{\theta}(\tau))\right\rangle$ is the energy, and the derivative in (15) acts only on the ket (right side). The matrix A is the **quantum metric tensor** (Fubini-Study metric), which captures the geometry of the parameter space.

Parameters are updated as:

$$\vec{\theta}(\tau + \Delta\tau) = \vec{\theta}(\tau) + \Delta\tau\, \dot{\vec{\theta}} \tag{16}$$

where $\Delta\tau$ acts as a learning rate.

5.3 The Ansatz Circuit (Fig. 2)

The variational ansatz uses 3 parameters on 2 qubits:

```
q0:  -Ry(θ0)-*------
                |
q1:  -Ry(θ1)-X-Ry(θ2)-
```

With 3 parameters, this circuit can produce any real superposition of 4 basis states (3 free parameters = 4 real amplitudes − 1 normalization − 0 phases since all are real).

Starting from $|00\rangle$:

$$\begin{aligned}|\psi(\theta_0, \theta_1, \theta_2)\rangle = \cos\tfrac{\theta_0}{2} \left(\cos\tfrac{\theta_1+\theta_2}{2} |00\rangle + \sin\tfrac{\theta_1+\theta_2}{2} |10\rangle\right) \\ + \sin\tfrac{\theta_0}{2} \left(\sin\tfrac{\theta_1-\theta_2}{2} |01\rangle + \cos\tfrac{\theta_1-\theta_2}{2} |11\rangle\right)\end{aligned} \tag{17}$$

(Note: state indices use Qiskit's $|q_1\, q_0\rangle$ ordering.)

5.4 Computing C_i and A_{ij}

Energy gradient C_i: We use the **parameter-shift rule** for RY gates:

$$\frac{\partial E}{\partial \theta_k} = \frac{E(\theta_k + \pi/2) - E(\theta_k - \pi/2)}{2} \tag{18}$$

This is exact (not a finite-difference approximation) because $RY(\theta) = e^{-i\theta Y/2}$ generates a sinusoidal parameter dependence.

Metric tensor A_{ij}: We use **finite differences** on the overlap function $f(\vec{\theta}) = \left\langle \psi_0 | \psi(\vec{\theta}) \middle| \psi_0 | \psi(\vec{\theta}) \right\rangle$:

$$A_{ij} \approx -\frac{f(\vec{\theta} + \varepsilon\hat{e}_i + \varepsilon\hat{e}_j) - f(\vec{\theta} + \varepsilon\hat{e}_i - \varepsilon\hat{e}_j) - f(\vec{\theta} - \varepsilon\hat{e}_i + \varepsilon\hat{e}_j) + f(\vec{\theta} - \varepsilon\hat{e}_i - \varepsilon\hat{e}_j)}{4\varepsilon^2} \tag{19}$$

with $\varepsilon = 10^{-4}$. The matrix A is regularized as $A + \lambda I$ ($\lambda = 10^{-4}$) to handle near-singular cases when the state is close to a critical point in parameter space.

5.5 Convergence

The ground state converges within ∼20–30 steps. Excited states (computed via the penalty method, Section 6) take slightly longer.

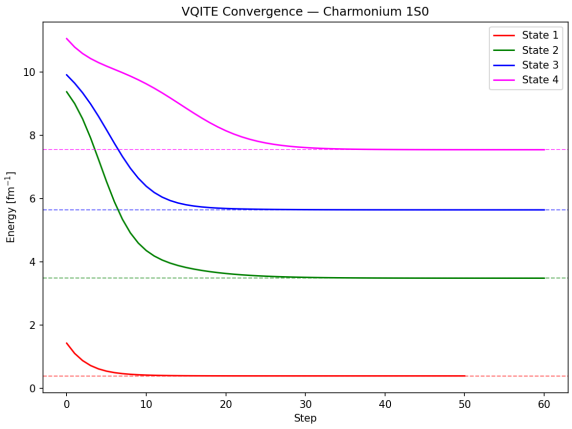
Code: `vqite.run_vqite()` implements the full loop. `vqite.compute_energy_gradient()` and `vqite.compute_metric_tensor()` compute C and A respectively.

6 Excited States

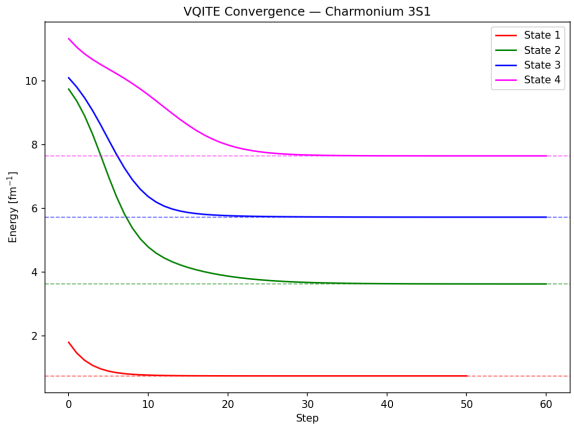
6.1 Penalty Method (Eq. 12)

To find the k-th excited state, we add a penalty term to the energy that pushes the variational state away from all lower-lying states:

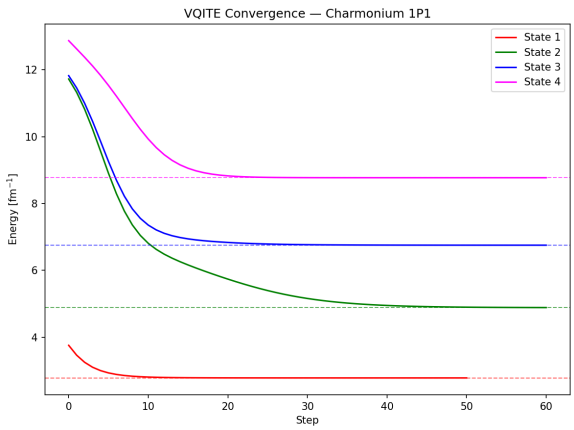
$$E_{\text{eff}} = \left\langle \psi(\vec{\theta}) \middle| H \middle| \psi(\vec{\theta}) \right\rangle + \alpha \sum_{j<k} \left| \left\langle \phi_j | \psi(\vec{\theta}) \middle| \phi_j | \psi(\vec{\theta}) \right\rangle \right|^2 \tag{20}$$



(a) $^1S_0$ channel



(b) $^3S_1$ channel



(c) $^1P_1$ channel

Figure 1: VQITE convergence for all three channels. Solid lines show the energy at each imaginary time step; horizontal dashed lines are the eigenvalues from diagonalizing the truncated Hamiltonian. Parameters: $\Delta\tau = 0.02$, initial $\theta_i = 0.5$, 50 steps. Compare with Figs. 4–6 of the paper.

where $|\phi_j\rangle$ are the previously determined eigenstates and α is a large positive coefficient.

Contrast with Gallimore: Gallimore uses an explicit orthogonalization in parameter space (reducing from 2 to 1 parameter for the first excited state). Woloshyn's penalty method is more general: it works for any number of excited states and any ansatz, at the cost of choosing α large enough.

In practice, $\alpha = 10$ works well for all channels. The overlap terms can be computed via the $U_i U_f^\dagger$ circuit (Fig. 3 of the paper) or directly from statevectors.

6.2 Sequential Computation

States are computed in order:

1. Ground state: VQITE with H
2. 1st excited: VQITE with $H + \alpha|\langle\phi_0|\cdot|\phi_0|\cdot\rangle|^2$
3. 2nd excited: VQITE with $H + \alpha\left(|\langle\phi_0|\cdot|\phi_0|\cdot\rangle|^2 + |\langle\phi_1|\cdot|\phi_1|\cdot\rangle|^2\right)$
4. 3rd excited: determined by the remaining orthogonal direction (or VQITE with 3 penalty terms)

Code: `vqite.run_vqite_excited()` takes a list of lower-state wavefunctions and adds the penalty.

7 M1 Transition Amplitudes

7.1 Physics

M1 transitions connect spin-singlet and spin-triplet states with the same orbital angular momentum (e.g., ${}^3S_1 \to {}^1S_0$). In the nonrelativistic limit, the transition involves a spin flip, and the spatial part of the amplitude is simply the wavefunction overlap:

$$\mathcal{A}_{M1} = \langle \psi_f | \psi_i | \psi_f | \psi_i \rangle \,, \qquad |\mathcal{A}_{M1}|^2 = \text{squared amplitude} \tag{21}$$

Because the spin-dependent interaction slightly distorts the spatial wavefunctions in different channels, the overlap is close to (but not exactly) 1 for corresponding states and close to 0 for non-corresponding states.

7.2 Quantum Circuits

Overlap circuit (Fig. 3): Prepare $|\psi_i\rangle$ then apply $U_f^\dagger$. Measuring $|00\rangle$ gives $P(00) = |\langle \psi_f | \psi_i | \psi_f | \psi_i \rangle|^2$.

Swap test (Fig. 7): Uses an ancilla qubit and controlled-SWAP. The measurement yields:

$$P(\text{ancilla} = 0) = \frac{1 + |\langle \psi_f | \psi_i | \psi_f | \psi_i \rangle|^2}{2} \tag{22}$$

This is less precise for small overlaps but doesn't require implementing $U^\dagger$.

7.3 Results

Transition	Paper (exact)	Our w.f. sim
$1^3S_1 \to 1^1S_0$	0.9826	0.994
$2^3S_1 \to 1^1S_0$	0.0107	0.004
$2^3S_1 \to 2^1S_0$	0.9781	0.994
$3^3S_1 \to 2^1S_0$	0.0061	0.002

The "exact" column uses solutions of the Schrödinger equation (Table 2 of the paper), while our results use the truncated basis. The dominant overlaps (~ 1) match well; the small off-diagonal overlaps differ because they are sensitive to basis truncation.

Code: `transitions.m1_overlap()` for statevector computation. `transitions.build_overlap_circuit()` and `transitions.build_swap_test_circuit()` for the quantum circuit versions.

8 E1 Transition Amplitudes

8.1 Physics

E1 transitions connect states with $\Delta L = 1$ (S-wave $\leftrightarrow$ P-wave) and the same spin. The transition amplitude is governed by the radial matrix element:

$$\langle u_f(r) | r | u_i(r) | u_f(r) | r | u_i(r) \rangle = \int_0^\infty u_f(r)\, r\, u_i(r)\, dr \tag{23}$$

where u_f is a P-wave ($l = 1$) state and u_i is an S-wave ($l = 0$) state.

8.2 Matrix Elements of r (Eq. 13)

The matrix $\langle m_P|r|n_S\rangle$ connecting P-wave and S-wave HO basis states is (in fm):

$$R = \begin{pmatrix} 0.578 & -0.472 & 0 & 0 \\ 0 & 0.746 & -0.667 & 0 \\ 0 & 0 & 0.882 & -0.817 \\ 0 & 0 & 0 & 1.000 \end{pmatrix} \tag{24}$$

The tridiagonal structure reflects the selection rule: HO states only couple to adjacent radial quantum numbers via r.

8.3 Pauli Decomposition (Eq. 14)

The r-operator matrix has **complex Pauli coefficients** (unlike the Hamiltonian which is real symmetric). The decomposition includes Y terms:

$$\hat{r} = c_0 + c_1 Z_0 + c_2 Z_1 + c_3 Z_0 Z_1 + c_4(X_0X_1 + Y_0Y_1) + c_5(X_1 - iY_1) + c_6 Z_0(X_1 + iY_1) + c_7(X_0Y_1 - Y_0X_1) \tag{25}$$

8.4 Hadamard Test (Fig. 8)

The Hadamard test circuit computes $\left\langle \psi_f|\hat{O}|\psi_i \middle| \psi_f|\hat{O}|\psi_i \right\rangle$ for an arbitrary operator $\hat{O}$. It uses an ancilla qubit, controlled state preparations, and a rotation R to select the real or imaginary part:

- $R = R_y(-\pi/2)$: ancilla measurement gives $\langle\sigma_x|\sigma_x\rangle = 2\,\mathrm{Re}\left\langle \psi_f|\hat{O}|\psi_i \middle| \psi_f|\hat{O}|\psi_i \right\rangle$
- $R = R_x(\pi/2)$: ancilla measurement gives $\langle\sigma_y|\sigma_y\rangle = 2\,\mathrm{Im}\left\langle \psi_f|\hat{O}|\psi_i \middle| \psi_f|\hat{O}|\psi_i \right\rangle$

Code: `transitions.e1_matrix_elements()` computes R. `transitions.e1_amplitude()` does statevector computation. `transitions.build_hadamard_test_circuit()` builds the circuit.

9 Readout Error Mitigation

9.1 The Problem

On real quantum hardware, qubits may be misidentified during measurement: $|0\rangle$ read as $|1\rangle$ and vice versa. For n qubits, the readout error is characterized by a $2^n \times 2^n$ calibration matrix M:

$$\vec{p}_{\text{noisy}} = M \cdot \vec{p}_{\text{ideal}} \tag{26}$$

where $\vec{p}$ is the vector of measurement probabilities.

9.2 Mitigation Strategy

The mitigation procedure (described in the paper's Section 5):

1. **Calibrate:** Prepare each computational basis state $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ and measure. The resulting statistics form the columns of M.
2. **Correct:** Apply M^{-1} to the measured probability vector to recover the ideal distribution.

For 2 qubits, M is only 4×4, making inversion straightforward. The paper uses the IBM `ibmq_manila` noise model for demonstration.

9.3 Effect on Results

The readout error mitigation is most impactful for:

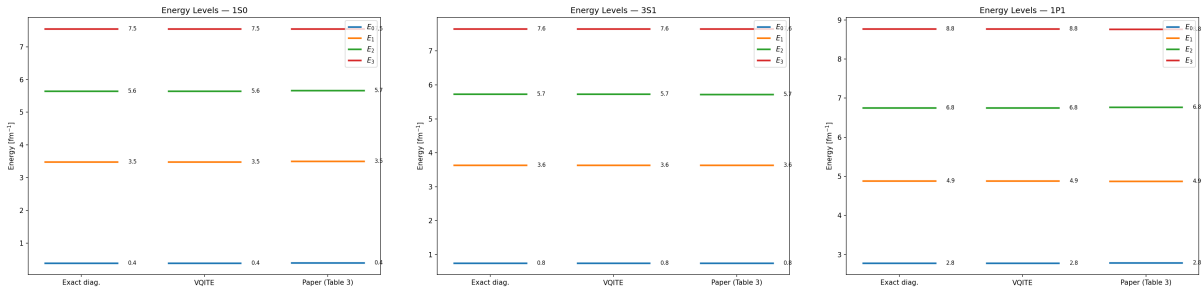
- **Energy calculations**: Figs. 9–11 show ideal vs. noisy vs. mitigated VQITE convergence. Mitigation brings noisy results close to ideal.
- **M1 overlaps**: Table 6 shows that large overlaps (~ 1) are well-recovered, but small overlaps (~ 0.001) remain unreliable even after mitigation.

10 Results Summary

10.1 Energy Levels

Our VQITE simulation reproduces the paper's eigenvalues (Table 3) to within the truncation error:

	$^1S_0$		$^3S_1$	
State	VQITE	Paper	VQITE	Paper
1	0.391	0.395	0.752	0.753
2	3.483	3.506	3.633	3.634
3	5.643	5.664	5.727	5.723
4	7.545	7.546	7.646	7.648



(a) $^1S_0$ (b) $^3S_1$ (c) $^1P_1$

Figure 2: Energy level comparison: VQITE results vs. exact diagonalization vs. paper's Table 3 values.

11 Code–Equation Mapping

Paper reference	Code location	Description
Eq. (1): Schrödinger eq.	`hamiltonian.compute_matrix()`	From-scratch computation
Eq. (3): H_{qm}	`hamiltonian.get_matrix()`	Paper's explicit matrices
Eq. (4): $^1S_0$ matrix	`PAPER_MATRICES["1S0"]`	Hardcoded constant
Section 2.2.1: Pauli form	`hamiltonian.matrix_to_pauli()`	$c_k = \text{Tr}(HP_k)/4$
Fig. 2: Ansatz circuit	`ansatz.build_ansatz()`	3-param Ry+CNOT
Eq. (9): VQITE update	`vqite.run_vqite()`	Full evolution loop
Eq. (10): Gradient C_i	`vqite.compute_energy_gradient()`	Parameter-shift rule
Eq. (11): Metric A_{ij}	`vqite.compute_metric_tensor()`	Finite differences
Eq. (12): Excited penalty	`vqite.run_vqite_excited()`	$E + \alpha \lvert \langle\phi\lvert\psi\rvert\phi\lvert\psi\rangle \rvert^2$
Eq. (13): $\langle m_P \lvert r \rvert n_S \rangle$	`transitions.e1_matrix_elements()`	Numerical integration
Eq. (14): r in Pauli form	`transitions.build_e1_pauli_operator()`	Complex coefficients
Fig. 3: Overlap circuit	`transitions.build_overlap_circuit()`	$U_i U_f^\dagger$
Fig. 7: Swap test	`transitions.build_swap_test_circuit()`	Ancilla + CSWAP
Fig. 8: Hadamard test	`transitions.build_hadamard_test_circuit()`	$\langle\psi_f\lvert O\rvert\psi_i\rvert\psi_f\lvert O\rvert\psi_i\rangle$

12 Running the Simulation

```
# Install
uv venv && source .venv/bin/activate
uv pip install -e .

# Run all channels
python -m quarksim.woloshyn.run --channel all --transitions

# Single channel, custom parameters
python -m quarksim.woloshyn.run --channel 1S0 --n-steps 80 --dtau 0.01

# Output directory
python -m quarksim.woloshyn.run --output results/woloshyn
```

Output files:

- `vqite_result_{channel}.json` — full simulation record per channel
- `vqite_convergence_{channel}.png` — energy vs. step (Fig. 1)
- `energy_levels_{channel}.png` — level comparison (Fig. 2)
- `wavefunction_{channel}.png` — ground state amplitudes